COUNTEREXAMPLES TO A CONJECTURE OF TSFASMAN AND VLADUT

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Abstract. In this note, we provide counterexamples to a conjecture of Tsfasman and Vladut concerning the support of infinite number fields. The proof relies on a recent result of Hajir, Larsen, Maire, and Ramakrishna.

1. Introduction

In the 1980s, Ihara initiated the asymptotic theory of global fields in the context of unramified Galois extensions of number fields [2]. Subsequently, Tsfasman and Vladut extended his work to arbitrary infinite global fields and introduced a collection of invariants associated with them [6]. In this note, we restrict our attention to the case of number fields. For a number field K, let $g_K := \frac{1}{2}\log|D_K|$ be its *genus* where D_K denotes its discriminant. For any prime power q, let $N_q(K)$ denote the number of primes of K of norm q, and let $N_{\mathbb{R}}(K)$ and $N_{\mathbb{C}}(K)$ denote the numbers of its real and complex primes, respectively. Recall that a sequence $\{K_i\}_{i\in\mathbb{N}}$ of number fields is called a *family* if K_i is not isomorphic to K_j for $i \neq j$. A family $\{K_i\}_{i\in\mathbb{N}}$ is called a *tower* if $K_i \subset K_{i+1}$. In any family, we have $g_i \to \infty$ as $i \to \infty$ because there are only finitely many number fields whose genus does not exceed g_0 for any real number g_0. Let A denote the set of parameters $\{p^k \mid p \text{ is a prime number},\ k \in \mathbb{N}^*\} \cup \{\mathbb{R}, \mathbb{C}\}$. We say that a family of number fields $\{K_i\}_{i\in\mathbb{N}}$ is *asymptotically exact* if the limit

$$\phi_q(\{K_i\}) := \lim_{i\to\infty} \frac{N_q(K_i)}{g_{K_i}}$$

exists for any $q \in A$. Tsfasman and Vladut proved that any tower of number fields is an asymptotically exact family and that the limit ϕ_q's depend only on $\bigcup_{i\in\mathbb{N}} K_i$. An *infinite number field* is the inductive limit of a tower of number fields:

$$\mathcal{K} := \bigcup_{i=1}^{\infty} K_i,$$

where $K_1 \subset K_2 \subset \cdots$. For an infinite number field $\mathcal{K} = \bigcup_{i=1}^{\infty} K_i$, one can define Tsfasman-Vladut invariants

$$\phi_q(\mathcal{K}) := \lim_{i\to\infty} \frac{N_q(K_i)}{g_{K_i}},$$

for $q \in A$. We define the *support* of an infinite number field $\mathcal{K}$ as

$$\mathrm{Supp}(\mathcal{K}) := \{q \in A \mid \phi_q(\mathcal{K}) > 0\}.$$

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An infinite profinite group is called *just-infinite* if all non-trivial closed normal subgroups have finite index. Inspired by Boston's work on the unramified Fontaine-Mazur conjecture, Tsfasman and Vladut proposed the following conjecture in their paper [6, Conjecture 6.1] (see also [5, Conjecture, Section 8.4.1]).

Conjecture 1.1. Let $\mathcal{K}$ be an infinite number field which is Galois over a number field K. Suppose that the corresponding Galois group $\mathrm{Gal}(\mathcal{K}/K)$ is just-infinite. Then the support $\mathrm{Supp}(\mathcal{K})$ of $\mathcal{K}$ is infinite.

In this note, our main result is the following.

Theorem 1.2. *Let K be a number field. Suppose that p is an odd prime number. Then there exists an infinite tamely ramified pro-p extension $\mathcal{K}$ of K satisfying the following:*

(1) *the Galois group $\mathrm{Gal}(\mathcal{K}/K)$ is just-infinite.*
(2) $\mathrm{Supp}(\mathcal{K}) = \emptyset$.

In particular, Conjecture 1.1 is false.

2. Proof of Theorem 1.2

If A is a commutative ring, then we denote by $\mathrm{SL}_n(A)$ the special linear group over A. Let p be a prime number. Let $\mathbb{Z}_p$ be the ring of p-adic integers and $\mathbb{F}_p$ the finite field of order p. For $n \geq 2$, we put

$$\mathrm{SL}_n^1(\mathbb{Z}_p) := \ker(\mathrm{SL}_n(\mathbb{Z}_p) \to \mathrm{SL}_n(\mathbb{F}_p)).$$

We have the following result.

Proposition 2.1. *Let p be an odd prime number and let $n \geq 2$ be an integer. Then the pro-p group $\mathrm{SL}_n^1(\mathbb{Z}_p)$ is just-infinite.*

Proof. Since p is odd, the Lie algebra $\mathfrak{sl}_n(\mathbb{F}_p)$ over the finite field $\mathbb{F}_p$ is simple. Then our claim follows from [4, Corollary 8.3]. □

Our argument relies crucially on the following recent result of Hajir, Larsen, Maire, and Ramakrishna.

Theorem 2.2. [1, Corollary C] *Let p be an odd prime number and let K be a number field. For $n \geq 2$, there exists an infinite tamely ramified pro-p extension $\mathcal{K}$ of K satisfying the following:*

(1) *$\mathcal{K}/K$ is ramified at infinitely many primes of K;*
(2) *the Galois group $\mathrm{Gal}(\mathcal{K}/K)$ is isomorphic to $\mathrm{SL}_n^1(\mathbb{Z}_p)$.*

We also need the following lemma.

Lemma 2.3. [3, Proposition 2.1] *Let K be a number field and $\mathcal{K}$ an infinite tamely ramified Galois extension of K. Suppose that $\mathcal{K}/K$ is ramified at infinitely many primes. Then $\mathrm{Supp}(\mathcal{K}) = \emptyset$.*

Now we can prove Theorem 1.2 as follows.

Proof of Theorem 1.2. By Theorem 2.2, there exists an infinite tamely ramified pro-p extension $\mathcal{K}$ of K satisfying the following:

(1) $\mathcal{K}/K$ is ramified at infinitely many primes of K;
(2) the Galois group $\mathrm{Gal}(\mathcal{K}/K)$ is isomorphic to $\mathrm{SL}_n^1(\mathbb{Z}_p)$.

By Proposition 2.1, the Galois group $\mathrm{Gal}(\mathcal{K}/K) \cong \mathrm{SL}_n^1(\mathbb{Z}_p)$ is just-infinite. By Lemma 2.3, we have $\mathrm{Supp}(\mathcal{K}) = \emptyset$. This completes the proof of our theorem. □

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